BAYESIAN MODELING OF COMPLEXITY

Agent-Based Modeling I: ODD Protocol

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Thursday 19th February, 2026

Hertie School of Public Governance

Agenda

1 Recap & Today's Goals

2 Modeling
- Modeling Fundamentals
- Agent-Based Modeling Concepts

3 The ODD Protocol
- Overview
- Design Concepts

4 Wrap-Up & Next Steps

1 Recap & Today's Goals

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Last Session: Complex Systems

Key takeaways:

- Complex systems exhibit **emergence**, **self-organization**, **adaptation**
- Feedback loops drive nonlinear dynamics
- Agent-based models (ABMs) capture bottom-up mechanisms

ABMs are useful when:

- Individual heterogeneity matters
- Local interactions drive system behavior
- Spatial/network structure is important
- We want to study emergent phenomena

This Session's Question

How do we actually build an ABM? We need a systematic framework.

Today's Learning Goals

By the end of this session, you will be able to:

1. **Understand** the ODD protocol for ABM design
2. **Write** ODD components for a given model
3. **Implement** basic ABM structure in Python (OOP)
4. **Begin coding** the Schelling Segregation Model:
   - Grid setup
   - Agent initialization
   - Basic class structure

Next session: Complete Schelling implementation + analysis

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What is a Model?

Definition

A model is a **purposeful representation** of some real system for solving a (category of) problem(s).

Keyword: Purposeful

- Models are simplifications, we keep what matters for our question
- Different questions → different models
- No single "correct" model

Simple models — Complex models

Learning the mechanisms — Predictive power

Analogy: Maps

Map type:	**Subway map**	**Topographic map**
Good for:	Navigation	Hiking, elevation
Bad for:	Distance estimation	Finding subway routes

Question determines fitness!

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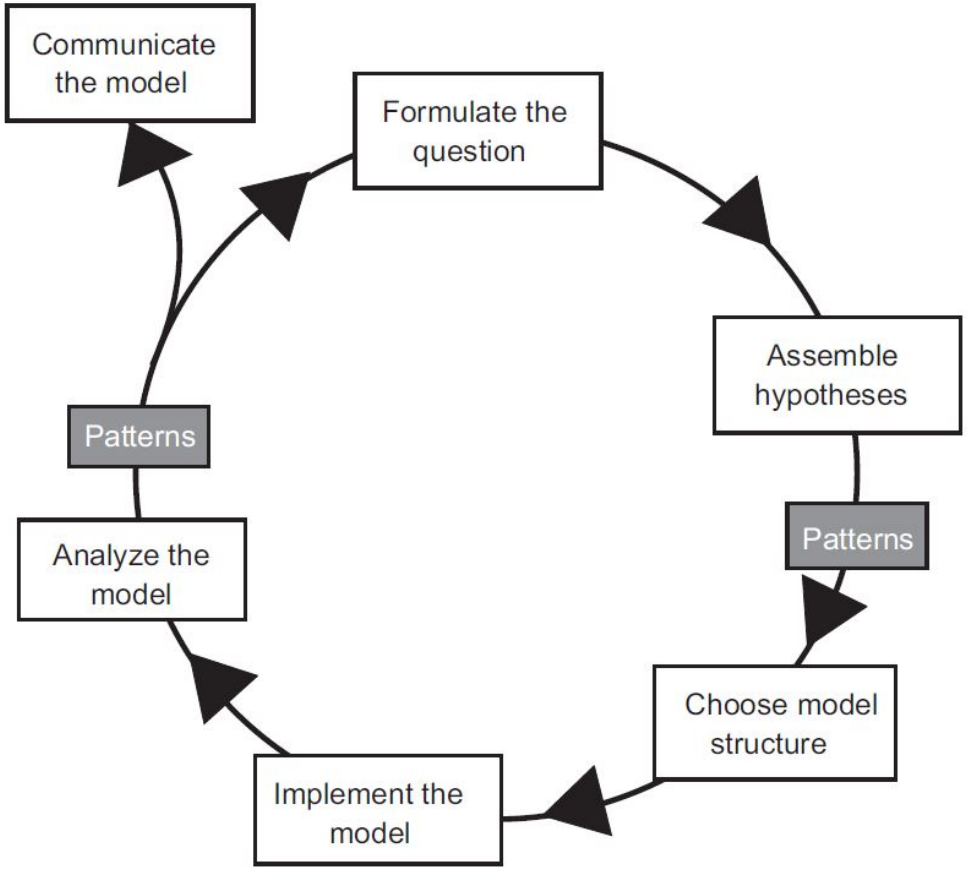
The Modeling Cycle



ABM development is iterative, not linear [1]

Key insight: We don't build models once, we refine through cycles of testing and revision

The Modeling Cycle: Five Steps

1. **Formulate the question**
   - Hardest step! In ABM, challenge is to stay focused
   - Be specific: "Why is there segregation?" is too broad
   - Better: "Can mild preferences produce strong segregation?"
2. **Assemble hypotheses for essential processes**
   - Start with **absolute minimalist** agents
   - What's the simplest mechanism that might work?
3. **Choose scales, entities, state variables, processes, parameters**
   - What are the agents? What environment?
   - What data do they track? What rules do they follow?
4. **Implement the model**
   - Write the code (we'll do this today!)
5. **Analyze, test, revise**
   - Does it behave as expected? Debug and iterate

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What Defines an ABM?

In ABM, we define **agents** (not aggregate state variables like in differential equations). **Agents are:**

- **Discrete entities:** Represent organisms, humans, firms, institutions, etc.
- **Unique:** Different from each other (location, size, preferences, memory)
- **Interactive:** Affect each other and environment, usually *locally*
- **Adaptive:** Change behavior based on environment, other agents, or own state

Why ABM?

ABMs allow us to address problems concerning **emergence**, i.e., system-level patterns that arise from individual-level interactions

Example: Traffic jams emerge from individual driving decisions, not from "jam-causing variables"

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Why Standardization Matters

The problem:

- ABMs can be extremely complex
- Hard to communicate design choices
- Difficult to reproduce others' work
- Easy to forget important details

The solution: ODD Protocol

- **O**verview, **D**esign concepts, **D**etails
- Developed by Grimm et al. (2006)
- Standard format for describing ABMs

Benefits

- Forces you to think through design systematically
- Makes models reproducible

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Overview, Design concepts, & Details (ODD) Protocol

	Elements of the ODD protocol
Overview	1. Purpose
	2. Entities, state variables, and scales
	3. Process overview and scheduling
Design concepts	4. Design concepts: ▪ Basic principles ▪ Emergence ▪ Adaptation ▪ Objectives ▪ Learning ▪ Prediction ▪ Sensing ▪ Interaction ▪ Stochasticity ▪ Collectives ▪ Observation
Details	5. Initialization
	6. Input data
	7. Submodels

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ODD I: Purpose and Patterns

Purpose:

- What is the system being modeled?
- What question does the model address?

Patterns (next session!):

- What real-world patterns should the model reproduce?
- Used for validation and calibration
- Example: "Model should produce 70-80% segregation"

Schelling Model - Purpose

Purpose: Explore whether mild individual preferences for same-type neighbors can produce strong residential segregation at the population level.
Pattern: Real cities show 70-90% segregation. Model should reproduce similar levels from individual tolerance thresholds.

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ODD II: Entities, State Variables, Scales

Entities: What are the agents and environment?

State variables: What information does each entity track?

Scales: What are the spatial and temporal extents?

Schelling Model: Entities& Variables

Entities:

- Agents: Households (two types: blue, red)
- Environment: 2D grid (patches/cells)

State Variables:

- *Agents:* type (blue/red), location (x, y), happiness (boolean)
- *Patches:* occupancy (empty/occupied)

Scales:

- Spatial: 50×50 grid, each cell = one household location
- Temporal: 1 time step = 1 opportunity to move

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ODD III: Process Overview & Scheduling

What happens in one time step? → Describe the sequence of events and who executes them

Schelling Model - Process Overview

Each time step:

- **For each agent** (in random order):
  - Calculate fraction of similar neighbors
  - If fraction < tolerance threshold → mark as unhappy
  - Else → mark as happy
- **For each unhappy agent** (in random order):
  - Find random empty cell
  - Move to that cell
- **Record metrics:**
  - % agents happy
  - Global segregation index
- **Stop if:** All agents happy OR 100 steps reached

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Scheduling: Why Order Matters

Question: Should agents move simultaneously or one at a time?

Two approaches:

1. **Synchronous updating:**
   - All agents assess happiness *then* all unhappy agents move
   - Environment state doesn't change during assessment phase
   - More "fair" → no first-mover advantage
2. **Asynchronous updating:**
   - Agents move immediately after deciding
   - Later agents see changed environment
   - Order matters → It can introduce bias or realism

Design Choice

In Schelling: Asynchronous (agents move in random order). This prevents coordination artifacts and is more realistic (people don't move simultaneously).

Always document your scheduling choice!

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Design Concepts: The "Why" Behind the Model

The Design Concepts section addresses **11 key questions** about model design:

1. Basic principles
2. Emergence
3. Adaptation
4. Objectives
5. Learning
6. Prediction
7. Sensing
8. Interaction
9. Stochasticity
10. Collectives
11. Observation

Not all apply to every model: Only describe what's relevant

Today: We'll cover the most important ones for Schelling and ABMs generally

Design Concepts: Basic Principles

Basic Principles: What general concepts, theories, or hypotheses inspire the model's design?

Key questions:

- What empirical/theoretical background motivated this model?
- What real-world phenomenon are we trying to understand?
- What existing theories or previous models does this build upon?

Schelling Model

Background: Urban residential segregation persists despite laws against discrimination

Theory: Schelling proposed that segregation can emerge from *mild* individual preferences, not necessarily from strong prejudice

Key insight: Small preference for "some" similar neighbors can cascade into extreme segregation at population level

Why state this explicitly?

- Connects the model to broader scientific/policy context
- Justifies modeling choices and assumptions

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Design Concept: Emergence

Emergence is the most fundamental concept in ABM

Key Questions

- What model dynamics arise in a complex, unpredictable way from agent behavior?
- What dynamics are imposed/predictable?

How to recognize emergence:

- Not simply the sum of individual properties
- Different *type* of result than individual properties
- Cannot easily be predicted from individuals' properties
- Usually takes time to appear

Schelling Example

Individual: Preference for $\geq$30% similar neighbors (mild)
Emergent: 80%+ segregation (strong)
Mild individual tolerance $\Rightarrow$ strong segregation (non-obvious!)

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Design Concept: Sensing

Sensing: What information do agents have access to?

Key questions:

- What type of information do agents use?
- How do they obtain it? (observe, communicate, remember?)
- Is information complete or partial? Accurate or biased?
- What is the scope/range of sensing?

Schelling Example

Agents sense:

- Types of neighbors in Moore neighborhood (8 adjacent cells)
- Information is perfect (no noise)
- Local only, i.e, agents don't know global segregation level

Design choice: Local sensing creates realistic **bounded rationality**

Realism: People typically don't have global information about cities when choosing where to live

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Design Concept: Adaptive Behavior & Objectives

Adaptation: Agents make decisions in response to changing conditions

Key questions:

- Do agents have objectives? (maximize utility, survive, reproduce?)
- How do agents make decisions toward these objectives?
- Are decisions optimal or "good enough" (**satisficing**)?

Schelling Example

Objective: Live in a neighborhood with sufficient same-type neighbors (meet tolerance threshold)

Decision rule:

- If current location meets threshold → stay (satisfied)
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Satisficing: Agents move to *any* empty cell, not the *best* one (computationally cheaper, more realistic → bounded rationality)

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Design Concept: Prediction

Prediction: Do agents form expectations about the future?

Decisions often depend on our notion of the future. Agents might:

- Develop statistical notions from historical data
- Use simulation history to update beliefs
- Employ **Bayesian inference** for belief updates

Example: Bayesian Agents (Advanced)

An agent starts with Beta-distributed prior belief about a parameter:

$$\theta \sim \text{Beta}(\alpha, \beta),$$

e.g., probability of business failure, infection risk, etc.
As agent observes outcomes (successes/failures), updates belief using Bayes' rule:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

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Design Concept: Interaction

Interaction: How do agents communicate or affect each other?

Types of interaction:

1. **Direct:** Agents affect each other's variables, decisions
   - Example: Information exchange, combat, trading
2. **Indirect:** Through shared environment/resources
   - Example: Competition for food, space

Schelling Example

Indirect interaction: Agents affect each other by occupying/vacating grid cells
When agent moves, it changes neighborhood composition for others $\rightarrow$ cascading effects
No direct communication between agents

Remark: When interaction is present, **order of execution matters**

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Design Concept: Stochasticity

Stochasticity: What processes involve randomness?

- Not all interesting dynamics result from stochasticity
- Often most interesting results come from **deterministic behavior in variable environment**

How to introduce stochasticity:

1. Choose distribution for variable of interest
2. Determine parameters and how they vary
3. Use replication (Monte Carlo) to assess impact

Purposes of stochasticity:

- Assign initial values (avoid artificial symmetry)
- Avoid modeling complex sub-processes (e.g., probability of rain instead of full weather model)

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Design Concept: Collectives

Collectives: Multiple agents linked in a specific way. **Examples:**

- Cells forming an organ
- Families or social groups
- Firms forming an industry

Two implementation approaches:

1. **Collectives as agents:**
   - Model group as single agent with own state variables
   - Less complex, faster computation
   - Example: Model households instead of individuals
2. **Emergent collectives:**
   - Represent as emergent characteristic of individual agents
   - More detailed, captures formation dynamics
   - Example: Social networks form from individual connections

Schelling: No explicit collectives (agents are independent households)

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   - Represent as emergent characteristic of individual agents
   - More detailed, captures formation dynamics
   - Example: Social networks form from individual connections

Schelling: No explicit collectives (agents are independent households)

Observation: Test & Control

Before analyzing results, we must verify the model's functionality. Three key testing approaches:

1. **Visualization**
   - Watch the model run (animation)
   - Check intermediate states
   - Does agent behavior look reasonable?
2. **Sanity checks**
   - Do outputs make sense?
   - Example: Population shouldn't exceed grid size
   - Conservation laws (if applicable)
3. **Stress tests (boundary conditions & edge cases)**
   - Extreme parameter values
   - Example: Tolerance = 0% should produce maximum segregation
   - Example: Tolerance = 100% should produce random mixing

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Why Visualize ABMs?

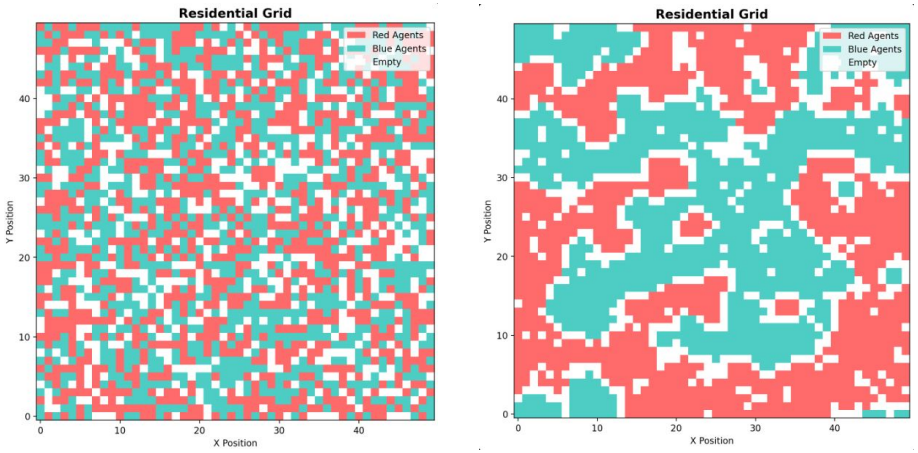
Visualization serves multiple purposes:

1. **Debugging & verification:**
   - Does agent behavior look correct?
   - Are spatial patterns forming as expected?
2. **Understanding dynamics:**
   - How do patterns emerge over time?
   - What's the trajectory to equilibrium?
3. **Communication:**
   - Show emergence to different audiences
4. **Exploration:**
   - Discover unexpected behaviors
   - Generate hypotheses for further testing

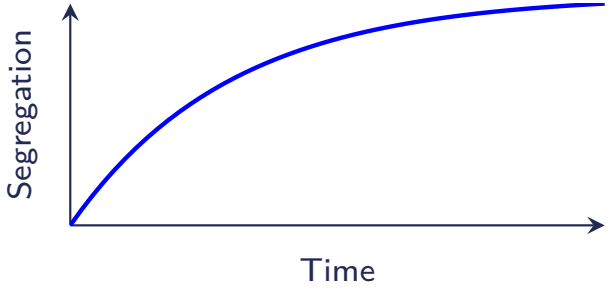
Types of Visualization for ABMs

1. **Spatial visualization:**
   - Grid snapshots (initial, final)
   - Animation over time
   - Spatial clustering metrics
2. **Time series:**
   - Metrics over time
   - Convergence plots
3. **Statistical summaries:**
   - Distributions
   - Histograms
   - Box plots

Live Demo



Left: first iteration, right: last iteration.



Time series

For Schelling: We'll focus on spatial (grid) + time series (metrics)

1 Recap & Today's Goals

2 Modeling

3 The ODD Protocol

4 Wrap-Up & Next Steps

What We Accomplished Today

- ✓ Understood the ODD protocol structure
- ✓ Learned key Design Concepts (emergence, sensing, adaptation, etc.)
- ✓ Wrote ODD Overview for Schelling model
- ✓ Implemented basic Schelling structure:
  - Grid class
  - Agent class
  - Model initialization

Partial model is working: We can create agents and place them on a grid

Still needed:

- Movement logic
- Simulation loop (time steps)
- Visualization
- Analysis & metrics

Next Session

Complete Schelling Implementation:

1. Implement movement rules
2. Add stopping conditions
3. Create visualization (animated grid)
4. Compute segregation metrics

Pattern-Oriented Modeling:

- What outputs should we track?
- How to validate against patterns?

Exploratory Analysis:

- Parameter sweeps (vary tolerance)
- Sensitivity analysis

ABM Project: Start thinking about systems you want to model!

Preparation

Before next session:

1. **Review today's code**
   - Make sure you understand each part
   - Try modifying parameters (grid size, density)
2. **Read:** Railsback & Grimm, Chapter 3-4 (ODD examples)
3. **Write:** ODD for a simple ABM of your choice
   - Can be hypothetical
   - Practice: Purpose, Entities, Process overview
4. **Think:** What complex system interests you for the project?

Questions?

Next session: Complete Schelling + Start Analysis